

SQL-Based Analysis of Single-Use Plastic Consumption and Environmental Awareness

Final Writing Report

A Mini-Project Based on Questionnaire Data

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Academic Project Report

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1. Introduction

The increasing use of single-use plastics has become an important environmental issue, as plastic waste affects both natural ecosystems and local communities. Products such as plastic bags, bottles, cups, straws, and cutlery are widely used in everyday life due to their convenience, low cost, and availability. However, their excessive use creates environmental concerns, especially when they are not reused, recycled, or properly disposed of.

The present mini-project focuses on the use of single-use plastics and the environmental awareness of citizens of Grevena and university students. The main purpose of the project is to collect, organize, and analyze real-world questionnaire data in order to examine participants' habits, attitudes, behavioral intentions, perceived barriers, and demographic characteristics related to single-use plastic consumption.

For this purpose, an anonymous academic questionnaire was designed and distributed to 20 participants. The questionnaire consists of 21 main questions and is divided into three parts. Part A focuses on the use of single-use plastics and reusable alternatives. Part B examines opinions, environmental attitudes, willingness to change behavior, and barriers to reducing plastic use. Part C includes general demographic information and consent for the future use of responses.

The project combines data collection, data coding, database design, and SQL analysis. Initially, the questionnaire responses were organized in Microsoft Excel through a structured data table, a variable dictionary, a coding guide, and lists of valid values. Afterwards, the data were transferred into a relational SQL database consisting of three main tables: Questions, Participants, and Responses. Through SQL queries, the project aims to extract useful results and identify patterns related to plastic use, environmental attitudes, behavioral intentions, and demographic differences.

Overall, this mini-project aims to demonstrate how real questionnaire data can be transformed into a structured database and analyzed through SQL in order to produce meaningful academic and research-oriented conclusions.

2. Research Aim and Objectives

The main aim of this mini-project is to investigate the use of single-use plastics, environmental attitudes, behavioral intentions, and perceived barriers among residents of Grevena and university students, using questionnaire data organized and analyzed through SQL.

More specifically, the project aims to examine how frequently participants use single-use plastic products, such as plastic bags, bottles, cups, straws, and cutlery. It also aims to explore the extent to which participants use reusable alternatives, such as reusable bags, reusable bottles, and reusable cups or containers.

Another objective of the project is to analyze participants' environmental attitudes toward single-use plastics and their willingness to change their daily habits in order to reduce plastic consumption. In addition, the project examines the main barriers that may prevent participants from reducing their use of single-use plastics, such as cost, habit, convenience, limited availability of alternatives, and lack of information.

Finally, the project aims to demonstrate how questionnaire data can be transformed into a relational database and analyzed through SQL queries. Through this process, the project connects data collection, coding, database design, validation, and analysis in order to produce meaningful research findings.

Research Objectives

The specific objectives of the project are:

1. To collect and organize questionnaire data related to single-use plastic consumption.
2. To code the responses using a clear numerical coding system.
3. To design a relational SQL database based on the questionnaire structure.
4. To validate the data by checking missing, invalid, and inconsistent responses.
5. To analyze the frequency of single-use plastic use and the use of reusable alternatives.
6. To examine participants' environmental attitudes and behavioral intentions.
7. To identify the strongest perceived barriers to reducing single-use plastic use.
8. To describe the demographic profile of the participants.
9. To compare selected findings with relevant international research.
10. To produce conclusions based on SQL analysis and questionnaire results.

3. Methodology

The methodology of this mini-project was based on a structured process that included questionnaire design, data collection, data coding, data organization in Excel, database creation, and SQL-based analysis. The aim was not only to collect questionnaire responses, but also to transform them into a structured dataset that could be stored, checked, and analyzed through a relational database.

The first step was the design of an anonymous academic questionnaire related to single-use plastic consumption and environmental awareness. The questionnaire consisted of 21 main questions and was divided into three parts. Part A focused on the use of single-use plastics and reusable alternatives. Part B examined environmental attitudes, behavioral intentions, perceptions, and barriers to reducing plastic use. Part C collected demographic information, such as gender, age group, educational level, employment status, place of residence, household size, and consent for future use of the responses.

The questionnaire was distributed to 20 participants, including residents of Grevena and university students. All responses were collected anonymously and were used only for academic and research purposes. Each participant was assigned a unique participant ID in order to organize the responses without using personal information.

After data collection, the responses were entered into Microsoft Excel. The Excel file was structured in a way that supported accurate data entry and documentation. It included a main data table, a variable dictionary, a coding guide, and lists of valid response values. Each answer was converted into a numerical code according to the predefined coding system. For example, scale-based questions were coded from 1 to 5, while demographic questions were coded according to their corresponding categories.

Special coding rules were also applied in order to handle missing and invalid responses. Missing or unanswered responses were recorded as NULL, while invalid multiple responses in single-choice questions were recorded with the code 99. During the SQL analysis, both NULL values and code 99 were excluded where necessary, especially when calculating averages and percentages.

The next step was the creation of a relational SQL database. The database was designed with three main tables: Questions, Participants, and Responses. The Questions table stored information about each questionnaire item, including the question code, question text, valid values, part, and group name. The Participants table stored the unique participant IDs. The Responses table stored the actual answers and worked as the central table linking participants with questions.

Finally, SQL queries were used to validate and analyze the data. Data validation queries checked the total number of questions, participants, and responses, as well as missing, invalid, and out-of-range values. Analysis queries were then used to calculate average scores, frequency distributions, percentages, group comparisons, and selected relationships between demographic characteristics and questionnaire responses. This methodology ensured that the project followed a clear and organized process from questionnaire design to final data analysis.

4. Database Design

The SQL database was designed as a small relational database in order to organize the questionnaire data in a clear and structured way. Instead of storing all answers in one wide table, the data were separated into three main tables: Questions, Participants, and Responses. This structure makes the database more flexible, easier to check, and more suitable for analysis.

The Questions table contains information about the questionnaire items. Each question has a unique QuestionID, a question code, the question text, the question type, valid values, the questionnaire part, and the thematic group to which it belongs. This table works as a reference table for the questionnaire structure and allows the analysis to be connected with the meaning of each question.

The Participants table contains the unique IDs of the 20 participants. Since the questionnaire was anonymous, no personal information was stored in the database. Each participant was represented only by a ParticipantID, which was used to connect the participant with their responses.

The Responses table is the central table of the database. Each row in this table represents one answer given by one participant to one question. The table includes a unique ResponseID, the ParticipantID, the QuestionID, and the coded answer value. The ParticipantID links each response to a participant, while the QuestionID links each response to the corresponding question.

The relationship between the tables is based on primary keys and foreign keys. The Participants table uses ParticipantID as its primary key, and the Questions table uses QuestionID as its primary key. In the Responses table, both ParticipantID and QuestionID are used as foreign keys. This means that every response must refer to an existing participant and an existing question.

The general structure of the database can be described as follows:

Participants → Responses ← Questions

This means that the Responses table connects the participants with the questions. One participant can have many responses, and one question can have many responses from different participants. This design follows the logic of a relational database and makes it possible to use SQL joins in order to display answers together with the corresponding question information.

For example, a join between the Responses and Questions tables can display each answer together with the question code and question text. This is important because the Responses table stores coded values, while the Questions table provides the meaning and context of each question. Therefore, the database design supports both efficient data storage and meaningful analysis.

5. Data Validation

Before proceeding to the final analysis, several SQL validation checks were performed in order to ensure that the database was correctly structured and that the data had been inserted properly. Data validation was an important step because the questionnaire data were manually transferred from Excel into the SQL database.

First, the total number of records in the main tables was checked. The `Questions` table contained 30 questionnaire variables, because questions 8, 9, and 10 were divided into four sub-questions each. The `Participants` table contained 20 participants, and the `Responses` table contained 600 responses in total. This result was expected, since 20 participants answered 30 coded questionnaire variables.

Additional checks were performed to confirm that each participant had 30 responses and that each question had 20 responses. This ensured that the data were complete in terms of structure and that no participant or question was accidentally omitted during the insertion process.

The validation process also included checks for missing and invalid responses. Missing answers were represented as `NULL`, while invalid multiple responses in single-choice questions were coded as 99. These values were identified through SQL queries and were excluded from calculations where necessary, especially in average score analysis.

A further validation query was used to detect possible values outside the expected numerical range. Since the questionnaire used coded values with maximum ranges up to 8, any value outside the acceptable range could indicate a data entry error. No out-of-range values were detected.

Finally, join queries were used to verify that the relational structure of the database worked correctly. By joining the `Responses` table with the `Questions` table through the common field `QuestionID`, it was possible to display each response together with the corresponding question code and question text. This confirmed that the foreign key relationships were functioning correctly and that the database was ready for analysis.

Overall, the validation process confirmed that the database contained the expected number of questions, participants, and responses, and that the data were suitable for SQL-based analysis.

6. ***SQL Analysis and Results***

The SQL analysis focused on extracting meaningful results from the questionnaire data. The main analysis included average scores, frequency distributions, group comparisons, and selected relationships between demographic characteristics and questionnaire responses. Missing values (NULL) and invalid responses (99) were excluded from the calculations where necessary.

The results are presented according to the main themes of the questionnaire: single-use plastic consumption, reusable alternatives, environmental attitudes, behavioral intentions, perceived barriers, perceptions and opinions, demographic profile, and selected comparative analyses.

6.1 Single-Use Plastic Use and Reusable Alternatives

The first part of the analysis examined the frequency of single-use plastic use and the use of reusable alternatives. The average frequency of plastic bag use was **3.45** on a 1–5 scale, indicating a moderate to relatively frequent use of plastic bags among the participants. The average use of reusable bags was **3.35**, suggesting that reusable bags are also used at a moderate level.

Among reusable alternatives, reusable bags had the highest average score (**M = 3.35**), followed by reusable bottles/flasks (**M = 3.05**). Reusable cups or utensils had the lowest average score (**M = 2.40**). This suggests that participants are more likely to use reusable bags and reusable bottles/flasks than reusable cups or containers in their daily life.

Overall, the results indicate that although participants show some use of reusable alternatives, single-use plastic products are still present in their daily habits. The relatively low score for reusable cups and utensils suggests that this type of reusable alternative has not yet been widely adopted within the sample.

6.2 Environmental Attitudes

Environmental attitudes were generally positive among the participants. The highest average score was found for the statement that single-use plastics significantly harm the environment (**M = 4.45**). This shows that participants strongly recognize the environmental burden caused by single-use plastics.

The belief that citizens can reduce plastic use through their daily behavior also received a high average score (**M = 4.35**). In addition, participants agreed that excessive plastic use is a problem for the local community of Grevena (**M = 4.17**). The lowest score in this group was related to being sufficiently informed about the effects of single-use plastics (**M = 3.75**), although this still indicates a relatively positive level of awareness.

These findings suggest that the participants have strong environmental concern and understand the negative impact of single-use plastic consumption. However, the slightly lower score regarding information indicates that there may still be room for further awareness and educational actions.

6.3 Behavioral Intentions

Behavioral intentions were also examined in order to understand whether participants were willing to change their daily habits and reduce single-use plastic consumption. The strongest behavioral intention was the intention to replace plastic bags with reusable bags, which had the highest average score (**M = 4.30**). This shows that participants were particularly willing to change their behavior regarding plastic bags.

The intention to use reusable cups or containers and the intention to reduce overall single-use plastic consumption both had an average score of **3.95**. These results indicate a generally positive willingness to adopt more environmentally friendly habits. The intention to replace plastic bottles with reusable bottles or flasks had a slightly lower average score (**M = 3.70**), but it still indicates a positive tendency toward behavioral change.

Overall, the findings suggest that participants are willing to reduce their use of single-use plastics, especially in relation to plastic bags. However, the difference between behavioral intentions and the actual use of some reusable alternatives suggests that positive intentions may not always lead directly to practical behavioral change.

6.4 Perceived Barriers

The analysis also examined the main barriers that may prevent participants from reducing their use of single-use plastics. The strongest barrier was habit and convenience, with an average score of **4.00**. This indicates that participants consider everyday habits and the convenience of single-use plastic products as the most important obstacle to reducing plastic consumption.

The second strongest barrier was lack of information or awareness, with an average score of **3.60**. This suggests that better information and awareness campaigns could help participants adopt more sustainable habits. The higher cost of alternative products had an average score of **3.40**, while limited availability of alternatives in stores had the lowest average score among the barriers (**M = 3.05**).

These findings show that reducing single-use plastic consumption is not only a matter of environmental awareness. Practical factors, such as habit, convenience, cost, availability, and information, also influence participants' ability to change their behavior. Therefore, successful reduction of single-use plastics requires both awareness and practical support for alternative choices.

6.5 Perceptions and Opinions

The analysis of perceptions and opinions provided additional information about how participants view the role of awareness, local willingness, and recommendations in reducing single-use plastic use. The highest average score in this section was found for the belief that information and awareness can help reduce plastic use (**M = 3.90**). This suggests that participants believe that informing citizens can contribute positively to reducing single-use plastic consumption.

The likelihood of recommending reusable products to others had an average score of **3.55**, indicating a moderately positive attitude toward encouraging other people to adopt reusable alternatives. The question related to the most helpful factor for reducing plastic use had an average score of **3.50**, based on 14 valid responses, as invalid multiple responses were excluded from the calculation.

The lowest score in this section was related to the perceived willingness of Grevena residents to change their habits, with an average score of **2.50**. This result suggests that participants may be skeptical about the readiness of the local community to reduce single-use plastic use. Overall, the findings show that although participants believe in the importance of awareness, they are less confident about the willingness of the wider local community to change its behavior.

6.6 Demographic Profile of Participants

The demographic analysis described the profile of the 20 participants who completed the questionnaire. In terms of gender, the sample consisted of 10 female participants (**50%**), 9 male participants (**45%**), and 1 participant who selected “Other” (**5%**). This shows that the sample was relatively balanced in terms of gender.

Regarding age group, the largest group was 18–24 years old, representing **55%** of the sample. The second largest age group was 25–34 years old (**20%**), followed by 45–54 years old (**15%**) and 65 and above (**10%**). No participants were recorded in the 35–44 and 55–64 age groups.

In terms of educational level, the largest group consisted of participants with higher education (**40%**). High school graduates and postgraduate or doctoral degree holders each represented **20%** of the sample. Participants with IEK or other technical education represented **15%**, while junior high school represented **5%**.

Regarding employment status, students formed the largest group (**40%**), followed by public sector employees (**30%**) and private sector employees (**20%**). Unemployed and retired participants each represented **5%** of the sample.

For place of residence, **70%** of participants selected “Other”, while **30%** reported living in Grevena Town. No participants selected a town or village in the Prefecture of Grevena. In terms of household size, the largest group consisted of households with three people (**40%**), followed by households with four or more people (**35%**). Finally, all participants gave consent for the future use of their responses, with **100%** answering “Yes”.

6.7 Comparative SQL Analysis

The final part of the SQL analysis examined selected relationships between demographic characteristics and questionnaire responses. These comparisons were included in order to show how the relational database structure can support more advanced analysis through SQL joins.

First, plastic bag use was compared across age groups. The highest average plastic bag use was found among participants aged 45–54 (**M = 3.67**), followed by participants aged 18–24 (**M = 3.55**) and participants aged 65 and above (**M = 3.50**). The lowest average was found among participants aged 25–34 (**M = 3.00**). These results suggest that plastic bag use differs slightly across age groups within the sample. However, because some age groups included a small number of participants, the results should be interpreted with caution.

Second, environmental attitudes were compared by educational level. Participants with higher education had the highest average environmental attitude score (**M = 4.35**), followed by junior high school participants (**M = 4.25**) and postgraduate or doctoral degree holders (**M = 4.19**). Overall, environmental attitudes were positive across all educational levels, suggesting that awareness of single-use plastic issues was generally high in the sample.

Third, behavioral intentions were compared by gender. Female participants had a slightly higher average behavioral intention score (**M = 4.00**) than male participants (**M = 3.92**). The “Other” gender category had the highest average score (**M = 4.25**), but this result should be interpreted carefully because it represented only one participant.

Overall, these comparative SQL analyses show that the database structure allowed the questionnaire responses to be examined in relation to demographic characteristics. The use of joins made it possible to connect demographic responses with behavioral and attitudinal variables, producing more detailed research findings.

7. Comparison with International Research

In order to strengthen the research dimension of the project, the findings of the present questionnaire were compared with international research related to single-use plastic reduction, environmental attitudes, and behavioral intentions. This comparison is useful because it places the local findings from Grevena and university students within a wider research context. For the international comparison, the present project refers to the study by Van et al. (2021), which examined factors affecting single-use plastic reduction behavioral intention. Van et al. (2021) found that attitudes play an important role in people's intention to reduce single-use plastic use. This finding is similar to the present project, where Environmental Attitudes had the highest group average (**M = 4.18**) and Behavioral Intentions also had a high average score (**M = 3.98**). Therefore, both studies suggest that environmental concern and positive attitudes are closely connected with the intention to reduce single-use plastic consumption.

International studies on single-use plastic consumption often show that environmental awareness and positive attitudes are important factors in encouraging people to reduce plastic use. However, these studies also highlight that positive attitudes do not always lead directly to behavioral change. Practical barriers, such as habit, convenience, cost, availability of alternatives, and lack of information, can prevent individuals from adopting more sustainable practices.

This is consistent with the findings of the present project. In this study, the highest group average was found for **Environmental Attitudes (M = 4.18)**, followed by **Behavioral Intentions (M = 3.98)**. These results suggest that participants were generally aware of the environmental problem of single-use plastics and showed willingness to reduce their use. However, the lowest group average was found for **Reusable Alternatives (M = 2.93)**, indicating that the actual use of reusable alternatives was lower than the level of awareness and intention.

A similar pattern is often discussed in international research, where a gap appears between what people intend to do and what they actually practice in everyday life (Van et al., 2021). In the present project, this gap is visible in the low average score for the use of reusable cups or utensils (**M = 2.40**), despite the high environmental concern expressed by participants. This suggests that practical habits and daily convenience may strongly influence behavior. However, the present project also showed that positive attitudes do not always lead to equally strong practical behavior. Although participants had high environmental attitudes and intentions, the use of Reusable Alternatives had the lowest group average (**M = 2.93**). This shows a possible gap between awareness, intention, and actual everyday behavior. This pattern is also consistent with Van et al. (2021), who examined behavioral intention as an important factor in single-use plastic reduction.

Another important finding of the present study was that **habit and convenience** formed the strongest perceived barrier to reducing single-use plastic use (**M = 4.00**). This supports the idea that plastic reduction is not only a matter of knowledge or environmental concern, but also depends on how easy, affordable, and practical alternative choices are in daily life. This finding shows that practical daily routines can limit sustainable behavior, even when people are environmentally aware. Compared

with the international study, this supports the idea that reducing single-use plastic consumption depends not only on attitudes, but also on practical conditions that make behavior change easier or more difficult.

Overall, the comparison with international research shows that the findings of this project follow a similar pattern: participants may have strong environmental awareness and positive intentions, but practical barriers can limit the adoption of reusable alternatives. Therefore, reducing single-use plastic consumption requires both awareness campaigns and practical measures that make reusable options easier and more convenient to use.

8. Conclusions

The present mini-project combined questionnaire data, Excel-based data organization, and SQL analysis in order to examine the use of single-use plastics and environmental awareness among residents of Grevena and university students. Through this process, the project demonstrated how real-world questionnaire responses can be transformed into a structured relational database and used for academic data analysis.

The results showed that participants generally had strong environmental attitudes toward single-use plastics. The highest scores were related to the perception that single-use plastics significantly harm the environment and that citizens can reduce plastic use through their daily behavior. This indicates that participants were aware of the environmental impact of single-use plastics and recognized the importance of individual and collective responsibility.

At the same time, the analysis showed that positive attitudes and intentions do not always lead to equally strong practical behavior. Although behavioral intentions were relatively high, especially regarding the replacement of plastic bags with reusable bags, the use of some reusable alternatives was lower. In particular, reusable cups and utensils had the lowest average score, suggesting that some sustainable alternatives are not yet widely adopted in everyday life.

The strongest barrier identified in the analysis was habit and convenience. This finding suggests that reducing single-use plastic use is not only a matter of environmental knowledge, but also depends on everyday routines, practical accessibility, and the convenience of alternative products. Lack of information and the cost of alternatives were also important barriers.

The demographic analysis showed that the sample included participants from different gender, age, education, and employment groups. The comparative SQL analysis also showed that the relational database structure made it possible to examine differences across demographic categories, such as age group, educational level, and gender.

Overall, the project achieved its main aim by combining data collection, coding, relational database design, validation, SQL analysis, and interpretation of results. The findings suggest that participants show high environmental awareness and positive intentions, but practical barriers may limit the consistent use of reusable alternatives.

Therefore, future actions aiming to reduce single-use plastic consumption should focus not only on awareness campaigns, but also on making reusable alternatives more accessible, affordable, and convenient in daily life.

9. References

Van, L., Abdul Hamid, N., Ahmad, M. F., Ahmad, A. N. A., Ruslan, R., & Muhamad Tamyaz, P. F. (2021). Factors of single use plastic reduction behavioral intention. *Emerging Science Journal*, 5(3), 269–278.
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10. Appendices

Appendix A: Questionnaire

This appendix refers to the academic questionnaire used for data collection. The questionnaire was anonymous and included questions about single-use plastic consumption, reusable alternatives, environmental attitudes, behavioral intentions, perceived barriers, demographic characteristics, and consent for future use of the responses.

File: *ACADEMIC QUESTIONNAIRE.docx*



QUESTIONNAIRE

Appendix B: Excel Dataset and Coding Structure

This appendix refers to the Microsoft Excel file used for the organization and coding of the questionnaire responses. The Excel file includes the main data table, variable dictionary, coding guide, and lists of valid response values.

File: *DATA of QUESTIONNAIRE.xlsx*



DATABASE

Appendix C: SQL Scripts

This appendix refers to the SQL files used for database creation, data insertion, validation, and analysis. The SQL scripts are organized into separate files in order to make the project structure clearer and more professional.

Files:

➤ *01_create_database_tables.sql*



create_database_tables.sql

➤ ***02_insert_questions_participants.sql***



insert_questions_participants.sql

➤ ***03_insert_responses.sql***



insert_responses.sql

➤ ***04_data_checks.sql***



data_checks.sql

➤ ***05_analysis_queries.sql***
questionnaire_project.sql



analysis_queries.sql

Appendix D: Project Topic

This appendix refers to the initial project topic document, which describes the purpose, structure, coding rules, and expected analysis of the mini-project.

File: *TITLE-TOPIC of PROJECT.docx*



TITLE-TOPIC

END OF THE REPORT